

Supplementary Material: Graph Characteristics

This supplementary document contains tables of descriptive statistics for graph characteristics across experimental factors.

Notation:

- $deg^+(n)$: out-degree of node n
- $\rho(G)$: density of graph G
- N_{trace} : set of nodes (events) selected for tracing
- R_{trace} : set of edges for given type of trace
- $G_{trace} = (N_{trace}, R_{trace})$: graph with the nodes and edges selected for tracing
- $Paths$: set of simple paths extracted by Algorithm 1.
- $N_{target} = \{n | n \in Entity \wedge type_n = type_{target}\}$: set of target entities

$|N_{trace}|$

Table S1: Descriptive statistics of $|N_{trace}|$ for S_{merge}

S_{merge}	count	mean	std	min	max
	50490	838.55	419.41	320.0	2064.0
DB	27600	761.18	247.87	432.0	1312.0
DB-FT	14800	782.82	229.99	496.0	1248.0
DB-Marking	16000	641.28	169.73	408.0	896.0
DB-Marking-FT	9690	695.88	140.61	504.0	880.0
DB-WB	16675	558.34	138.73	384.0	720.0
DB-WB-FT	9355	617.79	124.64	416.0	728.0
DB-WB-Marking	11010	560.51	131.27	372.0	704.0
DB-WB-Marking-FT	5860	657.54	47.94	612.0	708.0
FT	24370	1486.69	651.17	704.0	3776.0
Marking	24075	1228.01	521.12	528.0	2368.0
Marking-FT	13855	1206.56	472.18	720.0	2304.0
WB	24485	942.11	343.19	480.0	1664.0
WB-FT	13260	969.68	314.71	544.0	1600.0
WB-Marking	14425	823.54	257.72	456.0	1248.0
WB-Marking-FT	7900	878.36	205.04	680.0	1232.0
WT	26620	654.54	205.47	384.0	1136.0
WT-DB	20980	392.4	71.33	312.0	576.0
WT-DB-FT	11815	428.39	90.57	344.0	584.0
WT-DB-Marking	12955	380.44	98.37	300.0	560.0
WT-DB-Marking-FT	8020	447.25	107.65	348.0	564.0
WT-DB-WB	14780	341.0	91.03	288.0	536.0
WT-DB-WB-FT	7620	413.48	117.7	304.0	540.0
WT-DB-WB-Marking	9015	378.74	120.17	282.0	528.0
WT-DB-WB-Marking-FT	4450	530.0	0.0	530.0	530.0
WT-FT	14190	669.2	186.06	448.0	1072.0
WT-Marking	14810	525.72	124.16	360.0	720.0
WT-Marking-FT	8555	583.96	107.64	456.0	704.0
WT-WB	16160	446.27	94.24	336.0	624.0
WT-WB-FT	8995	492.85	101.21	368.0	632.0
WT-WB-Marking	10750	440.13	104.08	324.0	608.0
WT-WB-Marking-FT	4815	530.31	87.78	436.0	612.0

Table S2: Descriptive statistics of $|N_{trace}|$ for S_{split}

S_{split}	count	mean	std	min	max
	72355	679.88	150.47	320.0	1088.0
DB	28280	593.69	147.89	432.0	896.0
DB-FT	12970	575.57	130.98	440.0	832.0
DB-Marking	12540	725.84	211.14	512.0	1184.0
DB-Marking-FT	6300	700.71	130.95	544.0	896.0
DB-WB	12515	919.65	260.95	688.0	1376.0
DB-WB-FT	6090	874.92	201.59	720.0	1248.0
DB-WB-Marking	7000	1240.09	403.65	960.0	1952.0
DB-WB-Marking-FT	3505	1113.95	23.92	1088.0	1136.0
FT	42730	353.53	64.73	282.0	528.0
Marking	38245	401.34	94.61	300.0	704.0
Marking-FT	18005	406.0	78.96	308.0	560.0
WB	36995	475.24	126.34	344.0	800.0
WB-FT	17085	474.76	105.94	352.0	736.0
WB-Marking	17830	609.69	186.93	424.0	1088.0
WB-Marking-FT	9025	604.18	108.26	456.0	720.0
WT	30335	722.58	117.33	528.0	992.0
WT-DB	12990	1325.47	220.74	848.0	1664.0
WT-DB-FT	6575	1261.36	219.51	912.0	1536.0
WT-DB-Marking	6770	1616.39	396.61	1104.0	2240.0
WT-DB-Marking-FT	3150	1426.06	214.92	1232.0	1664.0
WT-DB-WB	6450	2093.68	500.57	1296.0	2624.0
WT-DB-WB-FT	3655	1890.19	472.03	1424.0	2368.0
WT-DB-WB-Marking	2575	2315.5	784.55	1808.0	3776.0
WT-DB-WB-Marking-FT	1710	2064.0	0.0	2064.0	2064.0
WT-FT	13365	714.47	109.2	560.0	928.0
WT-Marking	12730	856.83	190.01	656.0	1280.0
WT-Marking-FT	6505	827.92	115.8	720.0	992.0
WT-WB	13305	1045.84	239.32	752.0	1472.0
WT-WB-FT	6500	1004.54	196.33	816.0	1344.0
WT-WB-Marking	6815	1351.74	381.73	1008.0	2048.0
WT-WB-Marking-FT	3480	1228.05	87.92	1136.0	1312.0

Table S3: Descriptive statistics of $|N_{trace}|$ for D

D	count	mean	std	min	max
50	99520	752.89	443.36	282.0	3776.0
25-75	99985	758.24	431.91	288.0	3776.0
10-90	95835	714.99	396.84	282.0	3440.0
25-125	93775	706.88	378.07	282.0	2624.0
10-170	89265	707.58	412.11	282.0	3272.0

$|R_{trace}|$

Table S5: Descriptive statistics of $|R_{trace}|$ for S_{merge}

S_{merge}	count	mean	std	min	max
	50490	929.01	515.38	304.0	2553.0
DB	27600	824.5	305.84	416.0	1566.0
DB-FT	14800	820.84	264.81	496.0	1375.0
DB-Marking	16000	669.34	195.52	400.0	989.0
DB-Marking-FT	9690	725.44	156.87	512.0	943.0
DB-WB	16675	579.67	160.8	376.0	799.0
DB-WB-FT	9355	641.48	137.05	416.0	759.0
DB-WB-Marking	11010	581.26	145.8	368.0	735.0
DB-WB-Marking-FT	5860	687.17	47.39	640.0	739.0
FT	24370	1596.07	739.76	704.0	4254.0
Marking	24075	1328.32	608.58	518.0	2718.0
Marking-FT	13855	1277.02	533.57	736.0	2527.0
WB	24485	1018.74	408.54	464.0	1951.0
WB-FT	13260	1024.11	358.0	544.0	1759.0
WB-Marking	14425	867.66	295.01	448.0	1375.0
WB-Marking-FT	7900	921.55	231.55	704.0	1327.0
WT	26620	708.76	261.08	368.0	1373.0
WT-DB	20980	402.37	84.29	304.0	607.0
WT-DB-FT	11815	439.08	100.47	344.0	615.0
WT-DB-Marking	12955	388.67	109.92	296.0	591.0
WT-DB-Marking-FT	8020	463.69	120.13	352.0	595.0
WT-DB-WB	14780	345.64	102.35	284.0	567.0
WT-DB-WB-FT	7620	427.87	132.67	304.0	571.0
WT-DB-WB-Marking	9015	389.91	135.45	280.0	559.0
WT-DB-WB-Marking-FT	4450	560.68	0.47	560.0	561.0
WT-FT	14190	699.29	214.48	448.0	1183.0
WT-Marking	14810	543.89	140.74	352.0	794.0
WT-Marking-FT	8555	605.04	118.16	464.0	751.0
WT-WB	16160	457.71	107.22	328.0	655.0
WT-WB-FT	8995	508.77	112.3	368.0	663.0
WT-WB-Marking	10750	452.24	115.69	320.0	639.0
WT-WB-Marking-FT	4815	552.75	95.84	448.0	643.0

Table S7: Descriptive statistics of $|R_{trace}|$ for D

D	count	mean	std	min	max
50	99520	793.29	507.31	280.0	4253.0
25-75	99985	807.85	494.53	284.0	4254.0
10-90	95835	761.94	448.45	280.0	3821.0
25-125	93775	752.18	430.32	280.0	2845.0
10-170	89265	756.55	466.67	280.0	3606.0

Table S4: Descriptive statistics of $|N_{trace}|$ for r_f

r_f	count	mean	std	min	max
1	160975	802.54	434.92	282.0	2624.0
2	101980	760.34	439.58	282.0	3272.0
4	77870	687.43	410.25	282.0	3776.0
8	69710	645.55	354.8	282.0	3776.0
16	67845	640.46	342.53	282.0	3440.0

Table S6: Descriptive statistics of $|R_{trace}|$ for S_{split}

S_{split}	count	mean	std	min	max
	72355	706.56	150.77	304.0	1119.0
DB	28280	608.42	155.2	440.0	927.0
DB-FT	12970	604.85	148.01	448.0	894.0
DB-Marking	12540	765.58	230.48	517.0	1279.0
DB-Marking-FT	6300	786.44	157.08	576.0	1023.0
DB-WB	12515	970.9	287.25	704.0	1471.0
DB-WB-FT	6090	975.77	232.91	768.0	1407.0
DB-WB-Marking	7000	1367.67	450.34	1048.0	2174.0
DB-WB-Marking-FT	3505	1340.75	28.12	1304.0	1373.0
FT	42730	351.08	65.59	280.0	540.0
Marking	38245	402.1	97.1	296.0	735.0
Marking-FT	18005	419.01	91.84	304.0	604.0
WB	36995	481.49	132.5	344.0	830.0
WB-FT	17085	495.71	116.46	352.0	796.0
WB-Marking	17830	639.46	208.1	416.0	1182.0
WB-Marking-FT	9025	680.88	132.2	484.0	830.0
WT	30335	746.7	117.8	512.0	1022.0
WT-DB	12990	1409.26	237.47	864.0	1759.0
WT-DB-FT	6575	1394.03	241.8	992.0	1695.0
WT-DB-Marking	6770	1772.15	439.09	1188.0	2463.0
WT-DB-Marking-FT	3150	1679.37	241.54	1448.0	1951.0
WT-DB-WB	6450	2280.44	553.72	1380.0	2847.0
WT-DB-WB-FT	3655	2174.35	531.43	1639.0	2718.0
WT-DB-WB-Marking	2575	2592.03	866.85	2024.0	4254.0
WT-DB-WB-Marking-FT	1710	2523.54	36.02	2452.0	2553.0
WT-FT	13365	755.38	115.73	576.0	991.0
WT-Marking	12730	905.81	206.11	672.0	1374.0
WT-Marking-FT	6505	924.62	136.31	768.0	1119.0
WT-WB	13305	1110.06	259.52	768.0	1567.0
WT-WB-FT	6500	1112.34	223.39	896.0	1503.0
WT-WB-Marking	6815	1490.68	425.38	1096.0	2271.0
WT-WB-Marking-FT	3480	1467.43	97.16	1352.0	1566.0

Table S8: Descriptive statistics of $|R_{trace}|$ for r_f

r_f	count	mean	std	min	max
1	160975	856.92	495.68	280.0	2847.0
2	101980	810.1	498.9	280.0	3606.0
4	77870	728.83	467.82	280.0	4253.0
8	69710	682.25	404.29	280.0	4254.0
16	67845	677.13	391.31	280.0	3821.0

$$\rho(G_{trace})$$

Table S9: Descriptive statistics of $\rho(G_{trace})$ for S_{merge}

S_{merge}	count	mean	std	min	max
	50490	0.00157	0.00063	0.00058	0.00299
DB	27600	0.00154	0.00042	0.0009	0.00238
DB-FT	14800	0.00143	0.00037	0.00087	0.00208
DB-Marking	16000	0.00173	0.00043	0.0012	0.0025
DB-Marking-FT	9690	0.00156	0.00032	0.0012	0.00205
DB-WB	16675	0.00196	0.00045	0.00144	0.00266
DB-WB-FT	9355	0.00176	0.0004	0.00142	0.00245
DB-WB-Marking	11010	0.00196	0.00049	0.00147	0.00272
DB-WB-Marking-FT	5860	0.0016	0.00012	0.00147	0.00172
FT	24370	0.00084	0.0003	0.0003	0.00149
Marking	24075	0.00101	0.00037	0.00048	0.00194
Marking-FT	13855	0.00098	0.00029	0.00047	0.00145
WB	24485	0.00127	0.00041	0.0007	0.00215
WB-FT	13260	0.00119	0.00034	0.00068	0.00189
WB-Marking	14425	0.00139	0.00041	0.00088	0.00223
WB-Marking-FT	7900	0.00125	0.00024	0.00087	0.00154
WT	26620	0.00176	0.00043	0.00105	0.0027
WT-DB	20980	0.00267	0.00036	0.00179	0.00328
WT-DB-FT	11815	0.00248	0.00042	0.00179	0.00297
WT-DB-Marking	12955	0.00281	0.00053	0.00187	0.00338
WT-DB-Marking-FT	8020	0.00244	0.00052	0.00186	0.00294
WT-DB-WB	14780	0.00311	0.00054	0.00195	0.00352
WT-DB-WB-FT	7620	0.00268	0.00067	0.00195	0.00333
WT-DB-WB-Marking	9015	0.00294	0.00075	0.002	0.00357
WT-DB-WB-Marking-FT	4450	0.002	0.0	0.002	0.002
WT-FT	14190	0.00166	0.00038	0.00101	0.00231
WT-Marking	14810	0.00206	0.00044	0.00148	0.00284
WT-Marking-FT	8555	0.00183	0.00032	0.00149	0.00227
WT-WB	16160	0.00238	0.00043	0.00165	0.00305
WT-WB-FT	8995	0.00218	0.00041	0.00164	0.00278
WT-WB-Marking	10750	0.00245	0.0005	0.00172	0.00312
WT-WB-Marking-FT	4815	0.00202	0.00033	0.00171	0.00238

Table S10: Descriptive statistics of $\rho(G_{trace})$ for S_{split}

S_{split}	count	mean	std	min	max
	72355	0.00161	0.00036	0.00092	0.00298
DB	28280	0.00183	0.00041	0.00112	0.0024
DB-FT	12970	0.0019	0.00036	0.00127	0.00239
DB-Marking	12540	0.00156	0.00037	0.00089	0.00207
DB-Marking-FT	6300	0.00165	0.00028	0.00124	0.00205
DB-WB	12515	0.00123	0.00029	0.00077	0.00155
DB-WB-FT	6090	0.00133	0.00023	0.00089	0.00153
DB-WB-Marking	7000	0.00096	0.00023	0.00057	0.00116
DB-WB-Marking-FT	3505	0.00108	2e-05	0.00105	0.00112
FT	42730	0.0029	0.00048	0.00186	0.00357
Marking	38245	0.00262	0.00052	0.00142	0.00338
Marking-FT	18005	0.00262	0.00045	0.00185	0.00337
WB	36995	0.00226	0.00051	0.00127	0.00297
WB-FT	17085	0.00229	0.00044	0.00142	0.00296
WB-Marking	17830	0.00184	0.00044	0.00098	0.00248
WB-Marking-FT	9025	0.00192	0.00035	0.00155	0.00245
WT	30335	0.00147	0.00023	0.00101	0.00194
WT-DB	12990	0.00083	0.00016	0.00063	0.00124
WT-DB-FT	6575	0.00091	0.00018	0.00071	0.00123
WT-DB-Marking	6770	0.00072	0.00017	0.00049	0.001
WT-DB-Marking-FT	3150	0.00085	0.00013	0.0007	0.00097
WT-DB-WB	6450	0.00056	0.00016	0.00041	0.00084
WT-DB-WB-FT	3655	0.00065	0.00017	0.00048	0.00082
WT-DB-WB-Marking	2575	0.00053	0.00014	0.0003	0.00063
WT-DB-WB-Marking-FT	1710	0.00059	1e-05	0.00058	0.0006
WT-FT	13365	0.00152	0.00022	0.00112	0.00193
WT-Marking	12730	0.00129	0.00024	0.00083	0.00163
WT-Marking-FT	6505	0.00137	0.00018	0.00111	0.0016
WT-WB	13305	0.00106	0.00022	0.00072	0.00141
WT-WB-FT	6500	0.00114	0.00019	0.00082	0.00139
WT-WB-Marking	6815	0.00087	0.00019	0.00054	0.0011
WT-WB-Marking-FT	3480	0.00098	8e-05	0.0009	0.00106

Table S11: Descriptive statistics of $\rho(G_{trace})$ for D

D	count	mean	std	min	max
50	99520	0.00174	0.00075	0.0003	0.00353
25-75	99985	0.00172	0.0007	0.0003	0.00348
10-90	95835	0.00181	0.00074	0.00032	0.00357
25-125	93775	0.00183	0.00077	0.00041	0.00353
10-170	89265	0.00184	0.00073	0.00034	0.00357

Table S12: Descriptive statistics of $\rho(G_{trace})$ for r_f

r_f	count	mean	std	min	max
1	160975	0.00163	0.00069	0.00041	0.00353
2	101980	0.00173	0.00074	0.00034	0.00357
4	77870	0.00188	0.00075	0.0003	0.00353
8	69710	0.00195	0.00074	0.0003	0.00357
16	67845	0.00196	0.00074	0.00032	0.00353

$$\frac{\sum_{n \in N_{trace}} deg^+(n)}{|N_{trace}|}$$

Table S13: Descriptive statistics of $\frac{\sum_{n \in N_{trace}} deg^+(n)}{|N_{trace}|}$ for S_{merge}

S_{merge}	count	mean	std	min	max
DB	50490	1.08	0.07	0.95	1.24
DB-FT	27600	1.07	0.06	0.96	1.19
DB-Marking	14800	1.04	0.03	1.0	1.1
DB-Marking-FT	16000	1.04	0.03	0.98	1.1
DB-WB	9690	1.04	0.02	1.02	1.07
DB-WB-FT	16675	1.03	0.04	0.98	1.11
DB-WB-Marking	9355	1.04	0.02	1.0	1.07
DB-WB-Marking-FT	11010	1.03	0.03	0.99	1.06
FT	5860	1.05	0.0	1.04	1.05
Marking	24370	1.06	0.03	1.0	1.13
Marking-FT	24075	1.07	0.04	0.98	1.15
WB	13855	1.05	0.02	1.02	1.1
WB-FT	24485	1.07	0.05	0.97	1.17
WB-Marking	13260	1.05	0.03	1.0	1.1
WB-Marking-FT	14425	1.04	0.03	0.98	1.1
WT	7900	1.05	0.02	1.03	1.08
WT-DB	26620	1.07	0.06	0.96	1.21
WT-DB-FT	20980	1.02	0.04	0.97	1.12
WT-DB-Marking	11815	1.02	0.02	1.0	1.05
WT-DB-Marking-FT	12955	1.02	0.02	0.99	1.06
WT-DB-WB	8020	1.03	0.02	1.01	1.05
WT-DB-WB-FT	14780	1.01	0.02	0.99	1.06
WT-DB-WB-Marking	7620	1.03	0.03	1.0	1.06
WT-DB-WB-Marking-FT	9015	1.02	0.03	0.99	1.06
WT-FT	4450	1.06	0.0	1.06	1.06
WT-Marking	14190	1.04	0.03	1.0	1.1
WT-Marking-FT	14810	1.03	0.03	0.98	1.1
WT-WB	8555	1.03	0.01	1.02	1.07
WT-WB-FT	16160	1.02	0.03	0.98	1.11
WT-WB-Marking	8995	1.03	0.02	1.0	1.06
WT-WB-Marking-FT	10750	1.02	0.02	0.99	1.05
WT-WB-Marking-FT	4815	1.04	0.01	1.03	1.05

Table S14: Descriptive statistics of $\frac{\sum_{n \in N_{trace}} deg^+(n)}{|N_{trace}|}$ for S_{split}

S_{split}	count	mean	std	min	max
DB	72355	1.04	0.02	0.95	1.06
DB-FT	28280	1.02	0.02	0.97	1.05
DB-Marking	12970	1.05	0.02	1.0	1.09
DB-Marking-FT	12540	1.05	0.02	1.01	1.08
DB-WB	6300	1.12	0.03	1.06	1.16
DB-WB-FT	12515	1.05	0.02	1.02	1.08
DB-WB-Marking	6090	1.11	0.02	1.07	1.14
DB-WB-Marking-FT	7000	1.1	0.01	1.09	1.11
FT	3505	1.2	0.0	1.19	1.22
Marking	42730	0.99	0.01	0.95	1.03
Marking-FT	38245	1.0	0.02	0.96	1.04
WB	18005	1.03	0.03	0.98	1.1
WB-FT	36995	1.01	0.02	0.96	1.04
WB-Marking	17085	1.04	0.03	0.98	1.1
WB-Marking-FT	17830	1.04	0.02	0.98	1.09
WT	9025	1.12	0.02	1.06	1.17
WT-DB	30335	1.03	0.02	0.97	1.05
WT-DB-FT	12990	1.06	0.02	1.02	1.08
WT-DB-Marking	6575	1.11	0.01	1.09	1.12
WT-DB-Marking-FT	6770	1.1	0.01	1.08	1.1
WT-DB-WB	3150	1.18	0.01	1.17	1.19
WT-DB-WB-FT	6450	1.09	0.01	1.06	1.1
WT-DB-WB-Marking	3655	1.15	0.01	1.14	1.17
WT-DB-WB-Marking-FT	2575	1.12	0.01	1.1	1.13
WT-FT	1710	1.22	0.02	1.19	1.24
WT-Marking	13365	1.06	0.01	1.02	1.09
WT-Marking-FT	12730	1.06	0.01	1.02	1.07
WT-WB	6505	1.12	0.02	1.07	1.15
WT-WB-FT	13305	1.06	0.01	1.02	1.07
WT-WB-Marking	6500	1.11	0.01	1.09	1.13
WT-WB-Marking-FT	6815	1.1	0.01	1.09	1.11
WT-WB-Marking-FT	3480	1.2	0.01	1.18	1.21

Table S15: Descriptive statistics of $\frac{\sum_{n \in N_{trace}} deg^+(n)}{|N_{trace}|}$ for D

D	count	mean	std	min	max
50	99520	1.03	0.05	0.95	1.24
25-75	99985	1.05	0.05	0.95	1.24
10-90	95835	1.05	0.04	0.95	1.24
25-125	93775	1.05	0.05	0.95	1.22
10-170	89265	1.06	0.04	0.98	1.21

Table S16: Descriptive statistics of $\frac{\sum_{n \in N_{trace}} deg^+(n)}{|N_{trace}|}$ for r_f

r_f	count	mean	std	min	max
1	160975	1.05	0.05	0.95	1.23
2	101980	1.05	0.05	0.95	1.21
4	77870	1.04	0.05	0.95	1.24
8	69710	1.04	0.04	0.95	1.24
16	67845	1.04	0.05	0.95	1.24

$$\max(deg^+(n)|n \in N_{trace})$$

Table S17: Descriptive statistics of $\max(deg^+(n)|n \in N_{trace})$ for S_{merge}

S_{merge}	count	mean	std	min	max
	50490	6.77	5.62	2.0	40.0
DB	27600	6.67	4.95	2.0	40.0
DB-FT	14800	2.86	1.32	2.0	10.0
DB-Marking	16000	3.02	1.66	2.0	10.0
DB-Marking-FT	9690	2.05	0.22	2.0	3.0
DB-WB	16675	2.84	1.28	2.0	10.0
DB-WB-FT	9355	2.05	0.21	2.0	3.0
DB-WB-Marking	11010	2.04	0.19	2.0	3.0
DB-WB-Marking-FT	5860	2.0	0.0	2.0	2.0
FT	24370	6.01	3.64	2.0	22.0
Marking	24075	6.12	3.84	2.0	21.0
Marking-FT	13855	2.9	1.22	2.0	11.0
WB	24485	6.22	4.47	2.0	41.0
WB-FT	13260	2.87	1.06	2.0	8.0
WB-Marking	14425	2.79	0.87	2.0	6.0
WB-Marking-FT	7900	2.04	0.21	2.0	3.0
WT	26620	6.51	5.02	2.0	40.0
WT-DB	20980	3.18	1.89	2.0	10.0
WT-DB-FT	11815	2.07	0.26	2.0	3.0
WT-DB-Marking	12955	2.11	0.31	2.0	3.0
WT-DB-Marking-FT	8020	2.0	0.0	2.0	2.0
WT-DB-WB	14780	2.0	0.06	2.0	3.0
WT-DB-WB-FT	7620	2.0	0.0	2.0	2.0
WT-DB-WB-Marking	9015	2.0	0.0	2.0	2.0
WT-DB-WB-Marking-FT	4450	2.0	0.0	2.0	2.0
WT-FT	14190	2.84	1.19	2.0	10.0
WT-Marking	14810	2.74	1.1	2.0	7.0
WT-Marking-FT	8555	2.0	0.0	2.0	2.0
WT-WB	16160	2.72	1.39	2.0	10.0
WT-WB-FT	8995	2.06	0.23	2.0	3.0
WT-WB-Marking	10750	2.04	0.2	2.0	3.0
WT-WB-Marking-FT	4815	2.0	0.0	2.0	2.0

Table S18: Descriptive statistics of $\max(deg^+(n)|n \in N_{trace})$ for S_{split}

S_{split}	count	mean	std	min	max
	72355	2.19	0.54	2.0	5.0
DB	28280	2.36	0.7	2.0	5.0
DB-FT	12970	3.88	2.27	2.0	10.0
DB-Marking	12540	3.65	1.8	2.0	10.0
DB-Marking-FT	6300	7.54	4.06	2.0	20.0
DB-WB	12515	3.67	2.1	2.0	11.0
DB-WB-FT	6090	8.96	5.39	2.0	21.0
DB-WB-Marking	7000	6.79	2.78	3.0	21.0
DB-WB-Marking-FT	3505	14.03	5.85	9.0	40.0
FT	42730	2.24	0.71	2.0	5.0
Marking	38245	2.34	0.74	2.0	5.0
Marking-FT	18005	3.52	2.38	2.0	10.0
WB	36995	2.39	0.81	2.0	5.0
WB-FT	17085	3.94	2.37	2.0	10.0
WB-Marking	17830	3.58	1.96	2.0	11.0
WB-Marking-FT	9025	8.26	4.85	4.0	20.0
WT	30335	2.34	0.7	2.0	5.0
WT-DB	12990	3.48	1.58	2.0	11.0
WT-DB-FT	6575	7.05	3.21	3.0	18.0
WT-DB-Marking	6770	6.68	2.19	3.0	17.0
WT-DB-Marking-FT	3150	15.87	5.17	9.0	41.0
WT-DB-WB	6450	7.37	3.31	3.0	21.0
WT-DB-WB-FT	3655	11.16	3.27	9.0	27.0
WT-DB-WB-Marking	2575	12.32	4.44	9.0	33.0
WT-DB-WB-Marking-FT	1710	18.75	2.95	17.0	31.0
WT-FT	13365	3.77	2.0	2.0	10.0
WT-Marking	12730	3.73	1.97	2.0	11.0
WT-Marking-FT	6505	7.61	4.15	2.0	20.0
WT-WB	13305	3.93	1.79	2.0	11.0
WT-WB-FT	6500	6.8	2.95	4.0	20.0
WT-WB-Marking	6815	9.49	4.7	3.0	21.0
WT-WB-Marking-FT	3480	18.09	8.03	9.0	40.0

Table S19: Descriptive statistics of $\max(deg^+(n)|n \in N_{trace})$ for D

D	count	mean	std	min	max
50	99520	2.68	1.77	2.0	18.0
25-75	99985	3.33	2.52	2.0	18.0
10-90	95835	4.12	3.34	2.0	28.0
25-125	93775	3.82	3.17	2.0	18.0
10-170	89265	6.01	5.5	2.0	41.0

Table S20: Descriptive statistics of $\max(deg^+(n)|n \in N_{trace})$ for r_f

r_f	count	mean	std	min	max
1	160975	4.31	4.01	2.0	40.0
2	101980	4.04	3.62	2.0	33.0
4	77870	3.73	3.45	2.0	40.0
8	69710	3.54	3.07	2.0	40.0
16	67845	3.62	3.18	2.0	41.0

$|Paths|$

Table S21: Descriptive statistics of $|Paths|$ for S_{merge}

S_{merge}	count	mean	std	min	max
	50490	1.91	1.29	1.0	16.0
DB	27600	1.8	1.13	1.0	12.0
DB-FT	14800	1.37	0.72	1.0	8.0
DB-Marking	16000	1.4	0.77	1.0	8.0
DB-Marking-FT	9690	1.25	0.48	1.0	4.0
DB-WB	16675	1.27	0.6	1.0	4.0
DB-WB-FT	9355	1.32	0.5	1.0	4.0
DB-WB-Marking	11010	1.08	0.34	1.0	4.0
DB-WB-Marking-FT	5860	1.05	0.21	1.0	2.0
FT	24370	1.49	0.9	1.0	8.0
Marking	24075	1.74	1.18	1.0	12.0
Marking-FT	13855	1.26	0.56	1.0	6.0
WB	24485	1.74	1.15	1.0	16.0
WB-FT	13260	1.38	0.68	1.0	8.0
WB-Marking	14425	1.28	0.6	1.0	6.0
WB-Marking-FT	7900	1.15	0.45	1.0	4.0
WT	26620	1.82	1.28	1.0	16.0
WT-DB	20980	1.27	0.68	1.0	8.0
WT-DB-FT	11815	1.36	0.49	1.0	4.0
WT-DB-Marking	12955	1.28	0.55	1.0	4.0
WT-DB-Marking-FT	8020	1.29	0.46	1.0	2.0
WT-DB-WB	14780	1.05	0.24	1.0	3.0
WT-DB-WB-FT	7620	1.38	0.49	1.0	2.0
WT-DB-WB-Marking	9015	1.01	0.11	1.0	2.0
WT-DB-WB-Marking-FT	4450	1.0	0.0	1.0	1.0
WT-FT	14190	1.38	0.74	1.0	8.0
WT-Marking	14810	1.48	0.77	1.0	8.0
WT-Marking-FT	8555	1.26	0.52	1.0	4.0
WT-WB	16160	1.32	0.8	1.0	8.0
WT-WB-FT	8995	1.37	0.54	1.0	4.0
WT-WB-Marking	10750	1.13	0.37	1.0	4.0
WT-WB-Marking-FT	4815	1.09	0.28	1.0	2.0

Table S22: Descriptive statistics of $|Paths|$ for S_{split}

S_{split}	count	mean	std	min	max
	72355	1.0	0.0	1.0	1.0
DB	28280	1.1	0.29	1.0	2.0
DB-FT	12970	1.52	0.64	1.0	4.0
DB-Marking	12540	1.52	0.65	1.0	4.0
DB-Marking-FT	6300	2.66	1.45	1.0	8.0
DB-WB	12515	1.41	0.63	1.0	4.0
DB-WB-FT	6090	2.37	1.15	1.0	8.0
DB-WB-Marking	7000	1.85	1.21	1.0	8.0
DB-WB-Marking-FT	3505	3.44	2.43	1.0	16.0
FT	42730	1.13	0.34	1.0	2.0
Marking	38245	1.35	0.48	1.0	2.0
Marking-FT	18005	1.6	0.85	1.0	4.0
WB	36995	1.32	0.47	1.0	2.0
WB-FT	17085	1.8	0.86	1.0	4.0
WB-Marking	17830	1.42	0.69	1.0	4.0
WB-Marking-FT	9025	2.44	1.39	1.0	8.0
WT	30335	1.05	0.21	1.0	2.0
WT-DB	12990	1.18	0.52	1.0	4.0
WT-DB-FT	6575	2.04	0.95	1.0	8.0
WT-DB-Marking	6770	1.85	1.16	1.0	8.0
WT-DB-Marking-FT	3150	2.75	1.66	1.0	16.0
WT-DB-WB	6450	1.63	1.07	1.0	8.0
WT-DB-WB-FT	3655	2.99	1.87	1.0	12.0
WT-DB-WB-Marking	2575	1.99	1.39	1.0	8.0
WT-DB-WB-Marking-FT	1710	3.85	2.67	1.0	14.0
WT-FT	13365	1.48	0.59	1.0	4.0
WT-Marking	12730	1.48	0.58	1.0	4.0
WT-Marking-FT	6505	2.46	1.13	1.0	8.0
WT-WB	13305	1.38	0.59	1.0	4.0
WT-WB-FT	6500	2.37	1.17	1.0	8.0
WT-WB-Marking	6815	1.87	1.29	1.0	8.0
WT-WB-Marking-FT	3480	3.03	1.99	1.0	16.0

Table S23: Descriptive statistics of $|Paths|$ for D

D	count	mean	std	min	max
50	99520	1.42	0.85	1.0	12.0
25-75	99985	1.46	0.9	1.0	14.0
10-90	95835	1.45	0.89	1.0	14.0
25-125	93775	1.48	0.93	1.0	16.0
10-170	89265	1.49	0.91	1.0	16.0

Table S24: Descriptive statistics of $|Paths|$ for r_f

r_f	count	mean	std	min	max
1	160975	1.59	1.09	1.0	16.0
2	101980	1.46	0.89	1.0	16.0
4	77870	1.37	0.72	1.0	16.0
8	69710	1.36	0.69	1.0	9.0
16	67845	1.36	0.75	1.0	16.0

$$\frac{\sum_{p \in Paths} |p|}{|Paths|}$$

Table S25: Descriptive statistics of $\frac{\sum_{p \in Paths} |p|}{|Paths|}$ for S_{merge}

S_{merge}	count	mean	std	min	max
DB	50490	8.54	5.04	1.0	18.0
DB-FT	27600	8.76	5.24	1.0	18.0
DB-Marking	14800	9.2	5.2	2.0	18.0
DB-Marking-FT	16000	9.11	5.33	1.0	18.0
DB-WB	9690	9.6	5.32	2.0	18.0
DB-WB-FT	16675	9.03	5.42	1.0	18.0
DB-WB-Marking	9355	9.45	5.37	2.0	18.0
DB-WB-Marking-FT	11010	9.41	5.53	1.0	18.0
FT	5860	9.91	5.53	2.0	18.0
Marking	24370	8.98	4.97	2.0	18.0
Marking-FT	24075	8.9	5.17	1.0	18.0
WB	13855	9.39	5.12	2.0	18.0
WB-FT	24485	8.81	5.22	1.0	18.0
WB-Marking	13260	9.26	5.18	2.0	18.0
WB-Marking-FT	14425	9.2	5.33	1.0	18.0
WT	7900	9.7	5.31	2.0	18.0
WT-DB	26620	8.64	5.17	1.0	18.0
WT-DB-FT	20980	8.85	5.39	1.0	18.0
WT-DB-Marking	11815	9.32	5.35	2.0	18.0
WT-DB-Marking-FT	12955	9.23	5.47	1.0	18.0
WT-DB-WB	8020	9.72	5.48	2.0	18.0
WT-DB-WB-FT	14780	9.19	5.57	1.0	18.0
WT-DB-WB-Marking	7620	9.63	5.55	2.0	18.0
WT-DB-WB-Marking-FT	9015	9.61	5.68	1.0	18.0
WT-FT	4450	10.0	5.66	2.0	18.0
WT-Marking	14190	9.1	5.13	2.0	18.0
WT-Marking-FT	14810	8.99	5.24	1.0	18.0
WT-WB	8555	9.47	5.23	2.0	18.0
WT-WB-FT	16160	8.91	5.33	1.0	18.0
WT-WB-Marking	8995	9.39	5.31	2.0	18.0
WT-WB-Marking-FT	10750	9.34	5.46	1.0	18.0
WT-WB-Marking-FT	4815	9.79	5.44	2.0	18.0

Table S26: Descriptive statistics of $\frac{\sum_{p \in Paths} |p|}{|Paths|}$ for S_{split}

S_{split}	count	mean	std	min	max
DB	72355	8.69	5.12	1.0	18.0
DB-FT	28280	8.82	5.27	1.0	18.0
DB-Marking	12970	9.29	5.26	2.0	18.0
DB-Marking-FT	12540	9.13	5.36	1.0	18.0
DB-WB	6300	9.63	5.35	2.0	18.0
DB-WB-FT	12515	9.04	5.44	1.0	18.0
DB-WB-Marking	6090	9.48	5.39	2.0	18.0
DB-WB-Marking-FT	7000	9.4	5.55	1.0	18.0
FT	3505	9.91	5.53	2.0	18.0
Marking	42730	9.18	5.13	2.0	18.0
Marking-FT	38245	9.0	5.24	1.0	18.0
WB	18005	9.51	5.23	2.0	18.0
WB-FT	36995	8.93	5.3	1.0	18.0
WB-Marking	17085	9.35	5.25	2.0	18.0
WB-Marking-FT	17830	9.24	5.39	1.0	18.0
WT	9025	9.74	5.37	2.0	18.0
WT-DB	30335	8.76	5.23	1.0	18.0
WT-DB-FT	12990	8.85	5.37	1.0	18.0
WT-DB-Marking	6575	9.33	5.36	2.0	18.0
WT-DB-Marking-FT	6770	9.19	5.49	1.0	18.0
WT-DB-WB	3150	9.67	5.44	2.0	18.0
WT-DB-WB-FT	6450	9.09	5.56	1.0	18.0
WT-DB-WB-Marking	3655	9.6	5.53	2.0	18.0
WT-DB-WB-Marking-FT	2575	9.3	5.68	1.0	18.0
WT-FT	1710	10.0	5.66	2.0	18.0
WT-Marking	13365	9.2	5.2	2.0	18.0
WT-Marking-FT	12730	9.02	5.3	1.0	18.0
WT-WB	6505	9.54	5.29	2.0	18.0
WT-WB-FT	13305	8.96	5.38	1.0	18.0
WT-WB-Marking	6500	9.42	5.35	2.0	18.0
WT-WB-Marking-FT	6815	9.32	5.5	1.0	18.0
WT-WB-Marking-FT	3480	9.81	5.47	2.0	18.0

Table S27: Descriptive statistics of $\frac{\sum_{p \in Paths} |p|}{|Paths|}$ for D

D	count	mean	std	min	max
50	99520	9.1	5.29	1.0	18.0
25-75	99985	9.06	5.28	1.0	18.0
10-90	95835	9.07	5.29	1.0	18.0
25-125	93775	9.08	5.28	1.0	18.0
10-170	89265	9.08	5.29	1.0	18.0

Table S28: Descriptive statistics of $\frac{\sum_{p \in Paths} |p|}{|Paths|}$ for r_f

r_f	count	mean	std	min	max
1	160975	9.06	5.28	1.0	18.0
2	101980	9.09	5.29	1.0	18.0
4	77870	9.09	5.29	1.0	18.0
8	69710	9.08	5.29	1.0	18.0
16	67845	9.1	5.3	1.0	18.0

$$\max(\{|p| : p \in Paths\})$$

Table S29: Descriptive statistics of $\max(\{|p| : p \in Paths\})$ for S_{merge}

S_{merge}	count	mean	std	min	max
	50490	8.54	5.04	1.0	18.0
DB	27600	8.76	5.24	1.0	18.0
DB-FT	14800	9.2	5.2	2.0	18.0
DB-Marking	16000	9.11	5.33	1.0	18.0
DB-Marking-FT	9690	9.6	5.32	2.0	18.0
DB-WB	16675	9.03	5.42	1.0	18.0
DB-WB-FT	9355	9.45	5.37	2.0	18.0
DB-WB-Marking	11010	9.41	5.53	1.0	18.0
DB-WB-Marking-FT	5860	9.91	5.53	2.0	18.0
FT	24370	8.98	4.97	2.0	18.0
Marking	24075	8.9	5.17	1.0	18.0
Marking-FT	13855	9.39	5.12	2.0	18.0
WB	24485	8.81	5.22	1.0	18.0
WB-FT	13260	9.26	5.18	2.0	18.0
WB-Marking	14425	9.2	5.33	1.0	18.0
WB-Marking-FT	7900	9.7	5.31	2.0	18.0
WT	26620	8.64	5.17	1.0	18.0
WT-DB	20980	8.85	5.39	1.0	18.0
WT-DB-FT	11815	9.32	5.35	2.0	18.0
WT-DB-Marking	12955	9.23	5.47	1.0	18.0
WT-DB-Marking-FT	8020	9.72	5.48	2.0	18.0
WT-DB-WB	14780	9.19	5.57	1.0	18.0
WT-DB-WB-FT	7620	9.63	5.55	2.0	18.0
WT-DB-WB-Marking	9015	9.61	5.68	1.0	18.0
WT-DB-WB-Marking-FT	4450	10.0	5.66	2.0	18.0
WT-FT	14190	9.1	5.13	2.0	18.0
WT-Marking	14810	8.99	5.24	1.0	18.0
WT-Marking-FT	8555	9.47	5.23	2.0	18.0
WT-WB	16160	8.91	5.33	1.0	18.0
WT-WB-FT	8995	9.39	5.31	2.0	18.0
WT-WB-Marking	10750	9.34	5.46	1.0	18.0
WT-WB-Marking-FT	4815	9.79	5.44	2.0	18.0

Table S30: Descriptive statistics of $\max(\{|p| : p \in Paths\})$ for S_{split}

S_{split}	count	mean	std	min	max
	72355	8.69	5.12	1.0	18.0
DB	28280	8.82	5.27	1.0	18.0
DB-FT	12970	9.29	5.26	2.0	18.0
DB-Marking	12540	9.13	5.36	1.0	18.0
DB-Marking-FT	6300	9.63	5.35	2.0	18.0
DB-WB	12515	9.04	5.44	1.0	18.0
DB-WB-FT	6090	9.48	5.39	2.0	18.0
DB-WB-Marking	7000	9.4	5.55	1.0	18.0
DB-WB-Marking-FT	3505	9.91	5.53	2.0	18.0
FT	42730	9.18	5.13	2.0	18.0
Marking	38245	9.0	5.24	1.0	18.0
Marking-FT	18005	9.51	5.23	2.0	18.0
WB	36995	8.93	5.3	1.0	18.0
WB-FT	17085	9.35	5.25	2.0	18.0
WB-Marking	17830	9.24	5.39	1.0	18.0
WB-Marking-FT	9025	9.74	5.37	2.0	18.0
WT	30335	8.76	5.23	1.0	18.0
WT-DB	12990	8.85	5.37	1.0	18.0
WT-DB-FT	6575	9.33	5.36	2.0	18.0
WT-DB-Marking	6770	9.19	5.49	1.0	18.0
WT-DB-Marking-FT	3150	9.67	5.44	2.0	18.0
WT-DB-WB	6450	9.09	5.56	1.0	18.0
WT-DB-WB-FT	3655	9.6	5.53	2.0	18.0
WT-DB-WB-Marking	2575	9.3	5.68	1.0	18.0
WT-DB-WB-Marking-FT	1710	10.0	5.66	2.0	18.0
WT-FT	13365	9.2	5.2	2.0	18.0
WT-Marking	12730	9.02	5.3	1.0	18.0
WT-Marking-FT	6505	9.54	5.29	2.0	18.0
WT-WB	13305	8.96	5.38	1.0	18.0
WT-WB-FT	6500	9.42	5.35	2.0	18.0
WT-WB-Marking	6815	9.32	5.5	1.0	18.0
WT-WB-Marking-FT	3480	9.81	5.47	2.0	18.0

Table S31: Descriptive statistics of $\max(\{|p| : p \in Paths\})$ for D

D	count	mean	std	min	max
50	99520	9.1	5.29	1.0	18.0
25-75	99985	9.06	5.28	1.0	18.0
10-90	95835	9.07	5.29	1.0	18.0
25-125	93775	9.08	5.28	1.0	18.0
10-170	89265	9.08	5.29	1.0	18.0

Table S32: Descriptive statistics of $\max(\{|p| : p \in Paths\})$ for r_f

r_f	count	mean	std	min	max
1	160975	9.06	5.28	1.0	18.0
2	101980	9.09	5.29	1.0	18.0
4	77870	9.09	5.29	1.0	18.0
8	69710	9.08	5.29	1.0	18.0
16	67845	9.1	5.3	1.0	18.0

$|N_{target}|$

Table S33: Descriptive statistics of $|N_{target}|$ for S_{merge}

S_{merge}	count	mean	std	min	max
	50490	1.77	0.91	1.0	8.0
DB	27600	2.0	1.16	1.0	12.0
DB-FT	14800	2.29	1.35	1.0	11.0
DB-Marking	16000	2.11	1.26	1.0	10.0
DB-Marking-FT	9690	2.78	1.88	1.0	15.0
DB-WB	16675	2.12	1.34	1.0	12.0
DB-WB-FT	9355	2.62	1.93	1.0	15.0
DB-WB-Marking	11010	2.82	2.13	1.0	15.0
DB-WB-Marking-FT	5860	3.71	3.38	1.0	27.0
FT	24370	2.54	1.38	1.0	14.0
Marking	24075	2.22	1.29	1.0	13.0
Marking-FT	13855	2.58	1.54	1.0	14.0
WB	24485	2.09	1.26	1.0	12.0
WB-FT	13260	2.36	1.47	1.0	12.0
WB-Marking	14425	2.32	1.52	1.0	12.0
WB-Marking-FT	7900	2.96	2.32	1.0	16.0
WT	26620	1.79	1.0	1.0	11.0
WT-DB	20980	1.72	1.18	1.0	12.0
WT-DB-FT	11815	2.09	1.5	1.0	16.0
WT-DB-Marking	12955	2.16	1.7	1.0	16.0
WT-DB-Marking-FT	8020	2.92	2.59	1.0	26.0
WT-DB-WB	14780	2.33	1.9	1.0	16.0
WT-DB-WB-FT	7620	2.87	2.66	1.0	25.0
WT-DB-WB-Marking	9015	3.17	3.21	1.0	26.0
WT-DB-WB-Marking-FT	4450	4.32	5.11	1.0	32.0
WT-FT	14190	2.0	1.1	1.0	14.0
WT-Marking	14810	1.82	1.04	1.0	10.0
WT-Marking-FT	8555	2.39	1.61	1.0	16.0
WT-WB	16160	1.82	1.08	1.0	11.0
WT-WB-FT	8995	2.28	1.54	1.0	13.0
WT-WB-Marking	10750	2.43	1.78	1.0	15.0
WT-WB-Marking-FT	4815	3.2	2.79	1.0	16.0

Table S34: Descriptive statistics of $|N_{target}|$ for S_{split}

S_{split}	count	mean	std	min	max
	72355	2.66	2.56	1.0	32.0
DB	28280	2.2	1.61	1.0	16.0
DB-FT	12970	1.95	1.33	1.0	13.0
DB-Marking	12540	2.05	1.16	1.0	12.0
DB-Marking-FT	6300	1.86	0.99	1.0	11.0
DB-WB	12515	2.18	1.2	1.0	11.0
DB-WB-FT	6090	2.0	1.06	1.0	10.0
DB-WB-Marking	7000	2.65	1.25	1.0	14.0
DB-WB-Marking-FT	3505	2.43	1.09	1.0	8.0
FT	42730	2.14	1.78	1.0	16.0
Marking	38245	1.95	1.36	1.0	15.0
Marking-FT	18005	1.8	1.27	1.0	14.0
WB	36995	2.0	1.27	1.0	16.0
WB-FT	17085	1.68	0.97	1.0	10.0
WB-Marking	17830	2.03	1.12	1.0	11.0
WB-Marking-FT	9025	1.85	1.01	1.0	11.0
WT	30335	2.48	1.95	1.0	17.0
WT-DB	12990	2.5	1.75	1.0	16.0
WT-DB-FT	6575	2.25	1.45	1.0	12.0
WT-DB-Marking	6770	2.62	1.52	1.0	14.0
WT-DB-Marking-FT	3150	2.5	1.33	1.0	12.0
WT-DB-WB	6450	2.91	1.63	1.0	14.0
WT-DB-WB-FT	3655	2.63	1.3	1.0	11.0
WT-DB-WB-Marking	2575	3.47	1.71	1.0	13.0
WT-DB-WB-Marking-FT	1710	2.92	1.28	1.0	8.0
WT-FT	13365	2.13	1.5	1.0	13.0
WT-Marking	12730	2.21	1.33	1.0	12.0
WT-Marking-FT	6505	2.07	1.22	1.0	11.0
WT-WB	13305	2.44	1.34	1.0	12.0
WT-WB-FT	6500	2.2	1.26	1.0	10.0
WT-WB-Marking	6815	2.82	1.42	1.0	12.0
WT-WB-Marking-FT	3480	2.64	1.26	1.0	12.0

Table S35: Descriptive statistics of $|N_{target}|$ for D

D	count	mean	std	min	max
50	99520	2.15	1.61	1.0	16.0
25-75	99985	2.31	1.87	1.0	32.0
10-90	95835	2.29	1.7	1.0	26.0
25-125	93775	2.19	1.54	1.0	27.0
10-170	89265	2.3	1.7	1.0	25.0

Table S36: Descriptive statistics of $|N_{target}|$ for r_f

r_f	count	mean	std	min	max
1	160975	1.66	0.69	1.0	4.0
2	101980	2.14	1.14	1.0	8.0
4	77870	2.53	1.68	1.0	13.0
8	69710	2.79	2.28	1.0	16.0
16	67845	2.88	2.61	1.0	32.0